individuals (often referred to as the matrix). Examples include oases in the desert, cave systems, city parks in an urban environment, freshwater pools within a dry landscape, or fragments of mountain forest between nonforested valleys.

In the mid-20th century, ecologists began more intensive studies into species distribution on different islands, and how and why they varied. In the US, biologists Edward Wilson and Robert MacArthur constructed the first mathematical model of the factors at play in island ecosystems and, in 1967, they outlined a new theory of island biogeography.

Their theory proposed that each island reflected a balance between the rate of new species arriving there and the rate at which existing species become extinct. For example, a habitable but relatively empty island would have a low extinction rate since there are fewer species to become extinct. When more species arrive, competition for limited resources increases. At a certain point, smaller populations will be outcompeted, and the rate of species extinction will rise. An equilibrium point occurs when the species immigration rate and the rate of those becoming extinct are equal; this may remain constant until a change occurs in either rate.

The theory also proposes that the rate of immigration depends on the distance from the mainland, or another island, and declines with increased distance. The area of an island is a further factor. The larger it is, the lower its rate of extinction, because if native species are pushed out of prime habitat by new immigrants, they have a better chance of finding an alternative, albeit imperfect (“suboptimal”) habitat. Larger islands are also likely to have a greater variety of habitats or microhabitats in which to accommodate new immigrants. A combination of variety and lower rates of extinction produces a greater species mix than on a small island—the “species-area effect.” The actual species in the mix will change over time, as a result of colonization and extinction, but will remain relatively diverse.